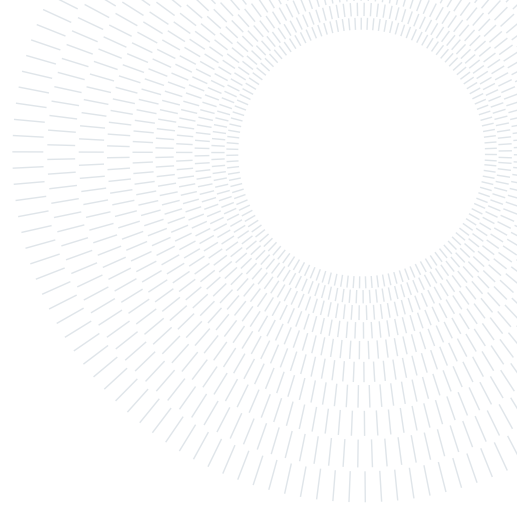




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Dividend Risk in an Autocallable Reverse Convertible

FINANCIAL ENGINEERING FINAL PROJECT

Group 5B: Stefano De Amici, Nassim Karimi, Alberto Toia

Advisors:

Roberto Baviera

Alessandro Montinari

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Abstract: We address the pricing, calibration, and risk management of a three-year Knock-In Reverse Convertible autocallable certificate on Intesa Sanpaolo. The note's value is driven by path-dependent digital-like triggers at the 70% knock-in barrier and the 100% autocall level. A flat Black-Scholes volatility would collapse the entire smile into a single number, mispricing the structure and making the prices inadequate. We therefore calibrate a semi-parametric local volatility surface with discrete proportional dividends through a fixed-point iteration on the Dupire forward PDE, and demonstrate analytically that Single Stock Dividend Futures have zero spot and volatility sensitivity under this framework, making them orthogonal instruments for pure dividend risk isolation. The risk profile is then managed by constructing four static hedging portfolios of increasing sophistication, with the final portfolio adopting an in-house method to select the optimal vanilla option that minimizes the L^2 residual of net Greeks across the spot grid.

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1. Introduction

This report presents the quantitative design, calibration, and validation of the pricing and risk engine constructed to manage a **short position on €20M** of product certificates written on **Intesa Sanpaolo (ISP)**.

This structured position exposes the trading book to a highly non-linear, multi-dimensional risk regime characterized by **short-volatility**, **short-dividend**, and **long-equity** exposures. In particular, the presence of a **down-and-in barrier at 70%** of the initial spot S_0 and periodic autocallable observation dates triggers **severe local gamma and vega spikes**, requiring a robust, tailored risk management framework.

To handle these path-dependent and skew-sensitive dynamics, the desk rejects flat-volatility Black-Scholes frameworks in favor of a **local volatility model** $\sigma_{LV}(S)$ calibrated via Fokker-Planck forward PDEs using the **Reghai fixed-point algorithm**.

Furthermore, we explicitly model cash dividend leakages using a **bootstrapped discrete proportional dividend framework**. In traditional models, dividends are assumed to be a fixed percentage (proportional yield) of the stock price, meaning that if the stock price drops, the expected cash dividend also drops. In reality, the market trades dividends via **Single Stock Dividend Futures (SSDFs)** at fixed cash amounts in EUR.

To resolve this discrepancy, our framework **dynamically adjusts the proportional dividend yield** so that the expected cash dividend is always locked to the fixed market price of the SSDF. Consequently, the dividend's cash value does not fluctuate with the stock price, allowing the desk to **hedge dividend risk in isolation using SSDFs**.

Phase	Quantitative Scope & Key Findings
Calibration & PDE Dynamics Section 3	We calibrate the local volatility surface $\sigma_{LV}(S)$ from discrete market smile quotes. We conduct a grid node sweep ($N \in [17, 71]$) to establish $N = 56$ as the optimal configuration that clamps residual calibration error below the 0.01% market tolerance, reducing the total Monte Carlo simulations required for Greek calculations by 90 runs (saving 4.5 million path simulations) without compromising calibration precision.
Par Pricing & Model Risk Section 5	We compute the fair coupon rate and certificate prices under both the calibrated Local Volatility model (LV-MC) and the flat ATM Black-Scholes model (BS-MC). We quantify the resulting <i>model risk</i> : collapsing the whole volatility skew onto a single ATM number misplices the certificate by 3.9% of notional and inflates the par coupon by roughly 110 bp (15.34% vs 14.24%).
Greeks & Dividend Proofs Section 6	We cross-validate the Greeks (Delta, Gamma, Vega, and Dividend Sensitivity) computed via finite differences utilizing Common Random Numbers against analytical vanilla benchmarks. We present the analytical proof demonstrating that SSDF Delta is exactly zero ($\Delta_{SSDF} \equiv 0$) under the bootstrapped proportional dividend model, proving its efficacy in isolating directional risk .
Hedge Portfolio Design Section 7	We design four distinct hedging portfolios (Portfolios A, B, C, and D) to neutralize the desk's short certificate book. We outline the logical progression from simple delta-replication to joint spot, dividend, and optimal vega profile hedging , where in Portfolio D we make use of a custom L^2 -greeks residual norm minimization algorithm.
P&L Explain & Attributions Section 9	We perform a valuation of the four portfolios across realistic spot and volatility paths and we formulate the P&L attribution equation to analyze how Portfolio D successfully minimizes residual P&L variance , confirming the desk's ability to immunize the book against joint market shocks.

2. Key Features & Hypotheses

Here we summarize the key modeling choices, discretization hypotheses, and structural assumptions underpinning our pricing and risk engine:

Feature / Hypothesis	Description	Rationale
Non-Uniform Grids	Both the Fokker-Planck PDE and Monte Carlo path generators use non-uniform grids that anchor key dates.	Prevents timing snap errors and interpolation bias at contractual event dates (dividend jumps, autocall tests, barrier observations).
5 Mandatory Grid Nodes	The LV grid strictly enforces 5 mandatory strikes: ATM (spot S_0 , which coincides with the 100% autocall level), ATMF (forward price), barrier (70%), and boundaries S_{low}, S_{high} .	Captures the non-linear contract payoffs and spatial grid boundaries exactly without interpolation noise or grid misalignment.
SSDF Orthogonality	Expected dividends are locked to market SSDF prices, resulting in zero Delta and zero Vega for SSDF contracts.	Decouples dividend risk from spot price movement and volatility smile shifts, allowing clean and orthogonal dividend hedging.
Common Random Numbers	Brownian path increments Z are shared exactly between the base and perturbed Monte Carlo paths.	Reduces Monte Carlo sampling noise, ensuring that finite-difference Greeks converge with low standard errors.
Central Differences	Greeks are computed via a central finite-difference scheme rather than one-sided forward differences.	Ensures a second-order truncation error ($\mathcal{O}(h^2)$) and prevents directional bias near the highly non-linear barrier and trigger levels.

Table 1: Key modeling features, discretization hypotheses, and hedging parameters of the pricing engine.

3. Calibration and Fokker–Planck PDE Dynamics

We calibrate a semi-parametric local volatility $\sigma_{LV}(S)$ to the market smile by repeatedly solving the *forward* (dual) Dupire equation for European call prices and matching model to market implied volatilities through a Reghai fixed-point iteration [?]. Consistently with the term sheet, all rates (discount, funding, repo) and dividend taxes are zero, and year fractions follow Act/365.

3.1. The forward Dupire / Fokker–Planck equation

With $r = q = 0$, the price of a European call $C(K, T)$ seen as a function of strike K and maturity T obeys the forward equation

$$\frac{\partial C}{\partial T} = \frac{1}{2} \sigma_{LV}^2(K) K^2 \frac{\partial^2 C}{\partial K^2}, \quad K \in (0, \infty), \quad T \in (0, T_{\max}], \quad (1)$$

with initial (today) condition $C(K, 0) = (S_0 - K)^+$. The local volatility is taken *time-independent* (semi-parametric in the strike dimension only) and piecewise-linear in log-strike between the calibration nodes, with linear extrapolation of the edge slopes outside the active market range (see Section 4.3).

3.2. Duffy change of variable and boundary conditions

To map the semi-infinite strike domain onto a bounded interval we apply the Duffy transform [? ?].

$$y = \frac{K}{K + S_0} \in [0, 1), \quad K = \frac{S_0 y}{1 - y}. \quad (2)$$

Under (2), equation (1) becomes a pure convection–diffusion equation in y ,

$$\frac{\partial C}{\partial T} = c(y) \frac{\partial C}{\partial y} + d(y) \frac{\partial^2 C}{\partial y^2}, \quad \begin{aligned} c(y) &= -\sigma_{LV}^2(y) y^2 (1 - y), \\ d(y) &= \frac{1}{2} \sigma_{LV}^2(y) y^2 (1 - y)^2. \end{aligned} \quad (3)$$

Both coefficients vanish at the endpoints ($y^2 \rightarrow 0$ as $y \rightarrow 0$ and $(1 - y) \rightarrow 0$ as $y \rightarrow 1$), so no flux reaches the boundary and Dirichlet conditions hold “for free”:

$$C(y=0, T) = S_0 \quad (\text{deep ITM}), \quad C(y \rightarrow 1, T) = 0 \quad (\text{deep OTM}). \quad (4)$$

3.3. Implicit finite-difference scheme

We discretize the transformed domain $y \in [0, 1)$ on a uniform spatial grid $y_i = i \Delta y$, $i = 0, \dots, N_y - 1$ with $N_y = 250$ grid points, and march in maturity using a fully implicit backward-Euler scheme with $N_t = 250$ uniform time steps, which is unconditionally stable [?]:

$$(I - \Delta t \mathcal{L}) \mathbf{C}^{n+1} = \mathbf{C}^n, \quad (5)$$

where $\mathcal{L}$ collects the central-difference approximations of ∂_y and ∂_{yy} in (3). The resulting operator is tridiagonal, with interior rows

$$\begin{aligned} A_{i,i} &= 1 + 2 \frac{\Delta t}{\Delta y^2} d(y_i), \\ A_{i,i-1} &= \frac{\Delta t}{2\Delta y} c(y_i) - \frac{\Delta t}{\Delta y^2} d(y_i), \\ A_{i,i+1} &= -\frac{\Delta t}{2\Delta y} c(y_i) - \frac{\Delta t}{\Delta y^2} d(y_i), \end{aligned} \quad (6)$$

Both the first and last rows of A are set to the identity ($A_{0,0} = A_{N_y-1,N_y-1} = 1$, off-diagonal entries zero); since the initial condition already satisfies $C_0^0 = S_0$ and $C_{N_y-1}^0 = 0$, the implicit step trivially propagates the Dirichlet boundaries. Crucially, since σ_{LV} is constant in time, A depends only on the step size Δt : it is assembled and LU-factorized *once* and reused at every step of equal Δt , which is the dominant computational saving of the solver.

3.4. Non-uniform time grid through ex-dividend dates

A uniform maturity partition would force each dividend jump to be applied at the nearest grid node $\text{round}(\tau_i/\Delta t)$, introducing an $\mathcal{O}(\Delta t)$ error in the dividend timing. We instead build a *non-uniform* time grid by inserting the

exact ex-dividend times into the uniform partition,

$$\{0 = t_0 < t_1 < \dots < t_N = T\} = \text{linspace}(0, T, N_t + 1) \cup \{\tau_1, \tau_2, \tau_3\}, \quad (7)$$

so that every dividend jump lands precisely on a grid node τ_i . The only cost is re-factorizing A whenever Δt changes (a handful of additional LU decompositions, cached across equal steps); the bulk of the time loop still reuses a single factorization.

3.5. Discrete proportional dividends

At each ex-dividend time τ_i a proportional dividend d_i detaches, $S(\tau_i^+) = (1 - d_i) S(\tau_i^-)$ [?]. The forward call price therefore satisfies the jump (continuity) condition

$$C(K, \tau_i^+) = (1 - d_i) C\left(\frac{K}{1 - d_i}, \tau_i^-\right), \quad (8)$$

which follows directly from $\mathbb{E}[(S(\tau_i^-)(1 - d_i) - K)^+] = (1 - d_i) \mathbb{E}[(S(\tau_i^-) - K/(1 - d_i))^+]$.

Numerically (8) is applied by interpolating the pre-jump solution *linearly* onto the rescaled strike grid $K/(1 - d_i)$. The PDE strike grid is dense enough to make linear accuracy sufficient: a higher-order scheme (such as a cubic spline, which would be the natural choice on sparser data) brings no precision benefit and risks introducing spurious oscillations.

3.6. Probability Density Evolution: Fokker–Planck Cascade

Before detailing the iterative calibration algorithm, we analyze the time evolution of the transition probability density function $p(S, t)$ of the underlying asset. By solving the forward Fokker-Planck (dual Dupire) equation starting from the Dirac delta initial condition $p(S, 0) = \delta(S - S_0)$ using the implicit finite-difference scheme described above, we obtain a visual map of the model's dynamics over the product's 3-year life.

Figure 1 illustrates this density cascade from three different viewing angles, providing a clear explanation of how the model behaves and what to expect from the density flow:

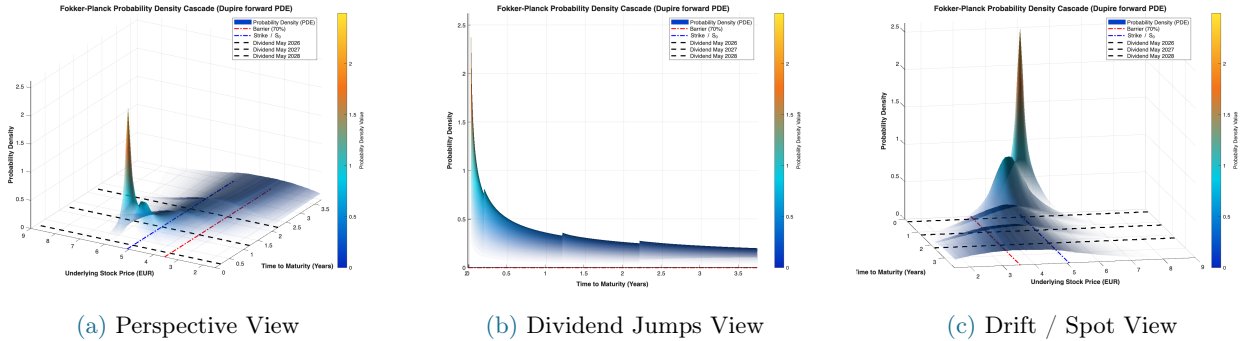


Figure 1: Fokker–Planck probability density cascade $p(S, t)$ from three viewing angles, solved under the calibrated Local Volatility model at t_0 .

1. **Perspective/General View** (Figure 1a): This perspective shows the global expansion of the probability density. The density starts as a sharp Dirac spike at $S_0 = 5.0740$ EUR at $t = 0$, and diffuses progressively over time. The widening of the distribution's base reflects the increasing variance of the stock price path over the 3-year horizon.
2. **Dividend Jumps View** (Figure 1b): Viewed from this angle, we observe the discrete ex-dividend dates (May 2026, May 2027, May 2028). At each of these dates, the proportional dividend payment d_i causes an instantaneous downward jump in the stock price: $S(\tau_i^+) = (1 - d_i) S(\tau_i^-)$. Numerically, this shifts the entire probability distribution to the left (towards lower spot values), creating sharp vertical "cut-offs" or step transitions in the density surface at τ_1, τ_2, τ_3 .
3. **Drift / Spot View** (Figure 1c): This frontal view along the time axis displays the progressive shift of the probability mass. Since the underlying asset pays cash dividends, its expected spot path undergoes a negative drift. Consequently, as maturity T approaches, the peak of the probability density drifts steadily to the left (downward in spot).

With the PDE dynamics visually clarified, we now present the iterative calibration algorithm used to align these dynamics with the market smile.

3.7. Reghai fixed-point calibration

The node volatilities $\{\sigma_j\}$ are calibrated by the multiplicative Reghai fixed point. Starting from a smile-based initial guess, each iteration k proceeds as follows:

1. solve (5)–(8) to obtain the model call prices $C^{(k)}(K)$ on the calibration strikes;
2. invert Black’s formula to get the model implied vols $IV_{\text{mod}}^{(k)}(K)$;
3. apply the multiplicative update

$$\sigma_j^{(k+1)} = \sigma_j^{(k)} \cdot \underbrace{\mathcal{I}_{\log K} \left[\frac{IV_{\text{mkt}}}{IV_{\text{mod}}^{(k)}} \right]}_{\text{ratio interpolated in } \log K} (K_j), \quad (9)$$

where the *ratio* (not σ itself) is interpolated in log-strike, which is smoother and cancels systematic bias when the nodes do not coincide with the market strikes where the *ratio* (not σ itself) is interpolated in log-strike, which is smoother and cancels systematic bias when the nodes do not coincide with the market strikes [?]..

The loop stops when $\max_K |IV_{\text{mkt}}(K) - IV_{\text{mod}}^{(k)}(K)| < \text{tol}$ (a 1 bp market tolerance). The operational workflow of this iterative fixed-point calibration is formalised as a Finite State Machine (FSM) in Figure 2.

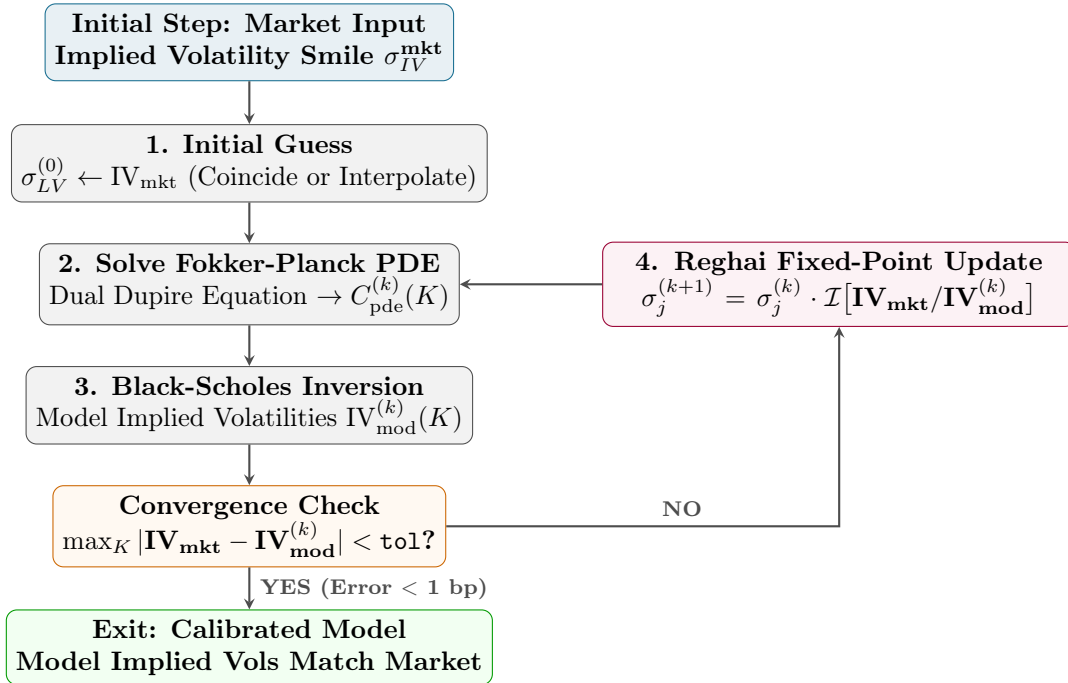


Figure 2: Reghai calibration FSM.

The final calibrated local volatility curve $\sigma_{LV}(S)$ is presented in Figure 3, showing the piecewise-linear profile obtained across strikes at inception.

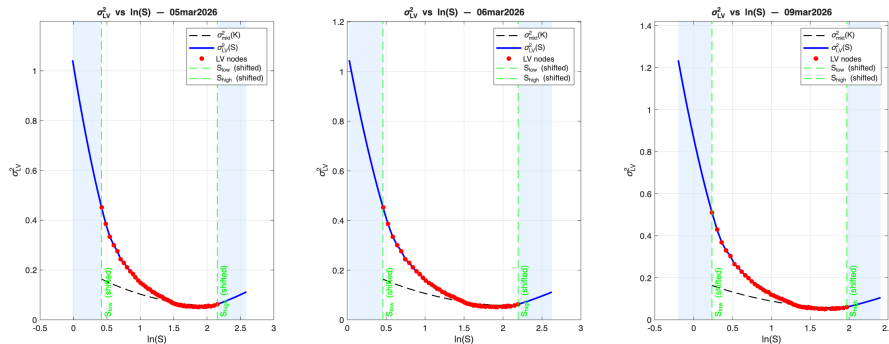


Figure 3: Calibrated local volatility profile $\sigma_{LV}(S)$ across strikes.

4. Choice of Nodes

We detail here the selection of the optimal grid size N for calibrating our Local Volatility (LV) model.

4.1. The Computational Challenge

The principal bottleneck of our pricing and risk engine lies in the computation of risk sensitivities (Greeks), particularly the search for the optimal Vanilla Put.

To mitigate the high computational cost associated with this bottleneck, we deployed the following computational optimizations:

- **Parallel Processing:** We utilized MATLAB's `parfor` constructs to calibrate the independent nodes and run Monte Carlo paths across multiple CPU cores.
- **Product-Specific Mandatory Nodes:** In all sparse grid configurations ($N < 71$), we strictly enforce that the critical contract strikes—namely ATM (spot S_0 , which coincides with the 100% autocall level), ATMF (forward price), the Barrier (70%), and the spatial boundaries (S_{low}, S_{high})—are included as mandatory grid nodes. This ensures that the highly non-linear regions of the Autocallable contract are perfectly captured without interpolation errors.

4.2. Trade-Off Analysis & Selection of $N = 56$

To determine the optimal number of nodes to include in the LV grid, we performed a sweep across the grid size $N \in [17, 20, 23, \dots, 71]$ and evaluated the residual calibration error (max difference between market and model implied volatilities), pricing impact, and CPU runtime. The empirical trade-offs are illustrated in Figure 4.

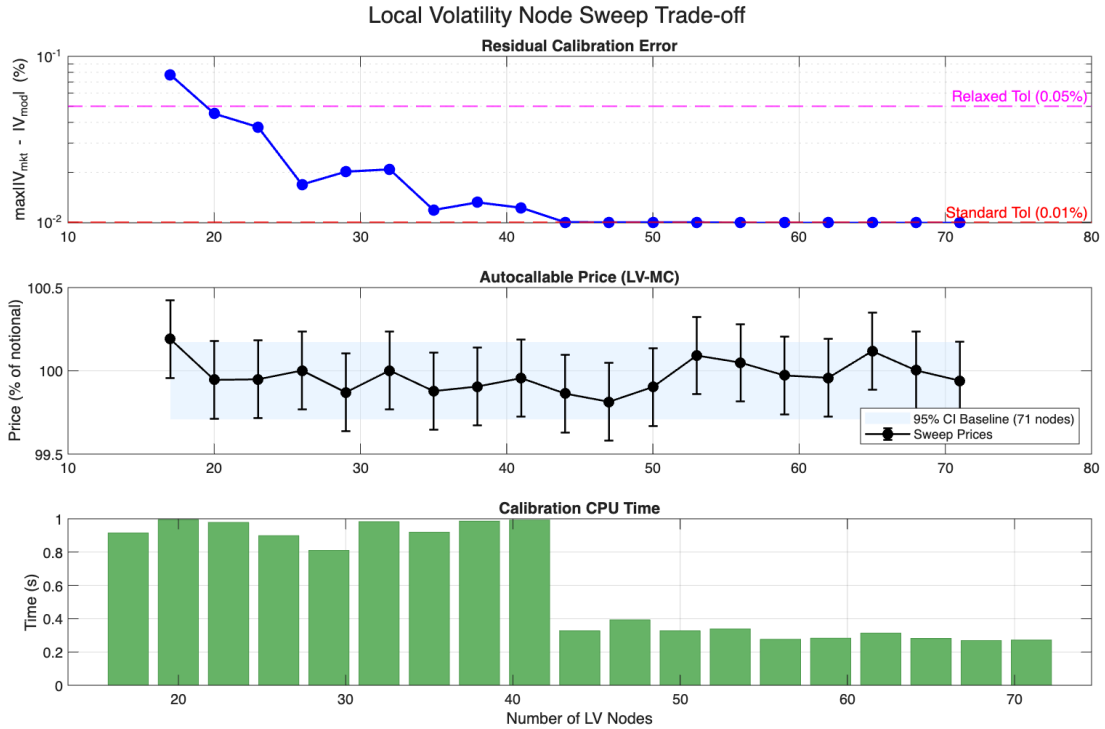


Figure 4: Trade-off analysis of Local Volatility node sweep. Top: Max calibration error ($|IV_{\text{mkt}} - IV_{\text{mod}}|$); Middle: Autocallable price (with 95% confidence interval); Bottom: Calibration CPU time.

The empirical analysis reveals a clear transition point:

1. **Calibration Error Floor & Entry Point:** Below $N = 44$ nodes, the residual calibration error exceeds the standard market tolerance of 0.01% (1 bp), requiring a relaxed tolerance threshold (up to 0.08%). At exactly $N = 44$, we reach the mathematical entry point where the calibration error is successfully clamped below the 0.01% threshold, making it the minimal precision-compliant and computationally advantageous configuration.
2. **Robustness Buffer & Safety Margin:** Although $N = 44$ satisfies the precision requirements, local volatility calibration is a numerical optimization process. To remain safe against localized grid discretization errors, potential interpolation noise, or stability issues near the highly non-linear contractual barriers

(70% coupon/principal barrier) and triggers (100% autocall), we opt for a conservative safety buffer. We select $N = 56$ nodes as a reasonably higher grid density that guarantees robust, smooth volatility gradients under all scenarios.

3. **Pricing Stability:** The middle panel confirms that for all N , the autocallable price remains exceptionally stable and well within the 95% confidence interval of the full 71-node baseline pricing.
4. **Computational Savings:** Choosing $N = 56$ instead of $N = 71$ reduces the number of Monte Carlo runs required for the Vega Bucket on each date from 142 to 112 (a 21.1% reduction). Across all three valuation dates (t_0, t_1, t_2) , this grid reduction reduces the total number of Monte Carlo runs in the pricing and risk-hedging pipeline from 617 to 527 runs. This saves 90 complete path simulations (equivalent to 4.5 million simulated paths of 50,000 paths each) while guaranteeing full, uncompromised calibration precision.

Consequently, we select $N = 56$ nodes as our optimal production configuration.

4.3. Domain Boundaries and Adaptive Grid Mapping

In the local volatility framework, defining the spatial grid boundaries $[S_{low}, S_{high}]$ is a crucial step for both the Fokker-Planck PDE solver and path simulations. Analytically, boundaries are established using a maturity-matched ATM volatility σ_{ATM} to define a 3-standard-deviation lognormal range around the forward price at maturity F_T :

$$hw = 3 \cdot \sigma_{ATM} \cdot \sqrt{T}, \quad S_{low}^{theory} = F_T \cdot e^{-hw}, \quad S_{high}^{theory} = F_T \cdot e^{hw} \quad (10)$$

For the ISP trade date of March 5, 2026 ($S_0 = 5.074$, ex-dividend dates in May, $T = 3.74$ years), the bootstrapped forward is $F_T = 3.855$. Utilizing $\sigma_{ATM} = 25.5\%$, the theoretical boundaries evaluate to $S_{low}^{theory} \approx 0.88$ ($\ln S \approx -0.13$) and $S_{high}^{theory} \approx 16.92$ ($\ln S \approx 2.83$).

However, the market smile is restricted to moneyness points $M = K/S_0 \in [0.3, 1.7]$, limiting observable quotes to $[K_{min}, K_{max}] = [1.52, 8.63]$. To prevent arbitrary extrapolation prior to calibration, we implement an adaptive grid mapping strategy (`select_adaptive_nodes.m`) that projects the theoretical boundaries onto the nearest available calibration strikes using a nearest-neighbor search:

$$S_{low} = \arg \min_{K \in \mathcal{K}_{calib}} |S_{low}^{theory} - K| = K_{min}, \quad S_{high} = \arg \min_{K \in \mathcal{K}_{calib}} |S_{high}^{theory} - K| = K_{max} \quad (11)$$

To select the $N = 56$ grid points from the 71 candidate market strikes, we implement an adaptive curvature-proportional sampling strategy in `select_adaptive_nodes.m`. The algorithm first retains all contractually mandatory strikes: the ATM level (spot S_0 , which coincides with the 100% autocall trigger), the ATMF forward price, the Barrier (70%), and the spatial boundaries (S_{low}, S_{high}) . For the remaining free nodes, we evaluate the local curvature of the market implied variance $V = IV_{mkt}^2$ in log-strike space using a second difference:

```
% Curvature in log-strike space (based on implied variance V)
V = IV_mkt .^ 2;
curv = abs(diff(diff(V)));
curv = [0; curv; 0];
```

We then allocate the remaining grid nodes proportionally to this curvature. This effectively discards (deletes) the nodes in regions of low curvature where the volatility smile behaves linearly and intermediate points can be omitted without losing interpolation accuracy, while keeping a dense distribution of nodes in the steepest skew regions near the barrier and trigger strikes. Outside this active calibrated domain $[S_{low}, S_{high}]$, local volatility is continuously piecewise-linear extrapolated in log-strike:

$$\sigma_{LV}^2(S) = \begin{cases} a + b \ln(S) & S < S_{low} \\ f(\ln S) & S \in [S_{low}, S_{high}] \\ c + d \ln(S) & S > S_{high} \end{cases} \quad (12)$$

where the boundary slopes b and d are computed dynamically by MATLAB's `griddedInterpolant` to match the outermost active grid segments, guaranteeing analytical continuity and tail stability.

5. Par Pricing and Model Risk

This section prices the structured note that the whole project is built around. We first dissect the product exactly as written in the Term Sheet, then describe the Monte Carlo engine that prices it, validate that engine on European vanillas, and finally present the fair coupon and the certificate price under the calibrated Local Volatility (LV) model and a flat Black–Scholes (BS) benchmark. The model-risk reading of those numbers closes the section.

5.1. The product: a Knock-In Reverse Convertible autocallable

The note is a three-year *Knock-In Reverse Convertible* on ISP, issued at par ($\mathcal{N}$ = EUR 100,000 per security) on the Strike Date S_0 . All payoffs are driven by the underlying ratio

$$R_t = \frac{S_t}{S_0}, \quad S_0 = \text{ISP closing price on the Strike Date (5 Mar 2026)}, \quad (13)$$

observed on three annual dates $t_1 = 5 \text{ Mar } 2027$, $t_2 = 5 \text{ Mar } 2028$ and $t_3 = 5 \text{ Mar } 2029$ (maturity). Three mechanisms act on these dates.

Mechanism	Dates	Condition	Payout / Action
Memory “Snowball” Coupon	t_1, t_2, t_3	$R_{t_i} \geq 70\%$	Pays annual coupon $X\%$ plus any accrued unpaid coupons.
		$R_{t_i} < 70\%$	Coupon is stored in memory (accrues) and paid on first later date where $R_{t_i} \geq 70\%$.
Autocall (Early Settlement)	t_1, t_2	$R_{t_i} \geq 100\%$	Note redeems early at par (100% of notional) together with that date’s coupon; terminates.
Final Redemption	t_3 (Maturity)	$R_{t_3} \geq 70\%$	Note redeems at par (100% of notional).
		$R_{t_3} < 70\%$	Knock-in occurs; note redeems at $\mathcal{N} \cdot R_{t_3}$ (30% discontinuous drop).

Table 2: Key contractual mechanisms of the Knock-In Reverse Convertible autocallable certificate.

Key Risk Observation: Embedded Short Down-and-In Put Option

Expanding the Term Sheet payoff below the barrier:

$$\mathcal{N} \cdot \max(100\% - \max(100\% - R_{t_3}; 0); 0) = \mathcal{N} \cdot R_{t_3}$$

reveals that the investor is effectively short a **maturity-only knock-in put option** (strike S_0 , knock-in barrier $70\% \cdot S_0$, observed only at t_3). Below the barrier the redemption switches from par to $\mathcal{N} \cdot R_{t_3}$, creating a discontinuity in the payoff slope that generates severe local gamma and vega spikes as the spot approaches the 70% level.

Because ISP pays discrete proportional dividends, the forward price at each observation date lies materially below the spot, reducing the probability of early autocall ($R_{t_i} \geq 100\%$) and increasing the probability of barrier breach at maturity, which raises the value of the embedded short put and directly penalizes the certificate from the investor’s perspective.

Figure 5 illustrates these mechanisms on three representative Monte Carlo trajectories: an early autocall at t_1 , a path that skips a coupon and recovers it through the memory snowball, and a path that breaches the 70% barrier at maturity and triggers the knock-in loss.

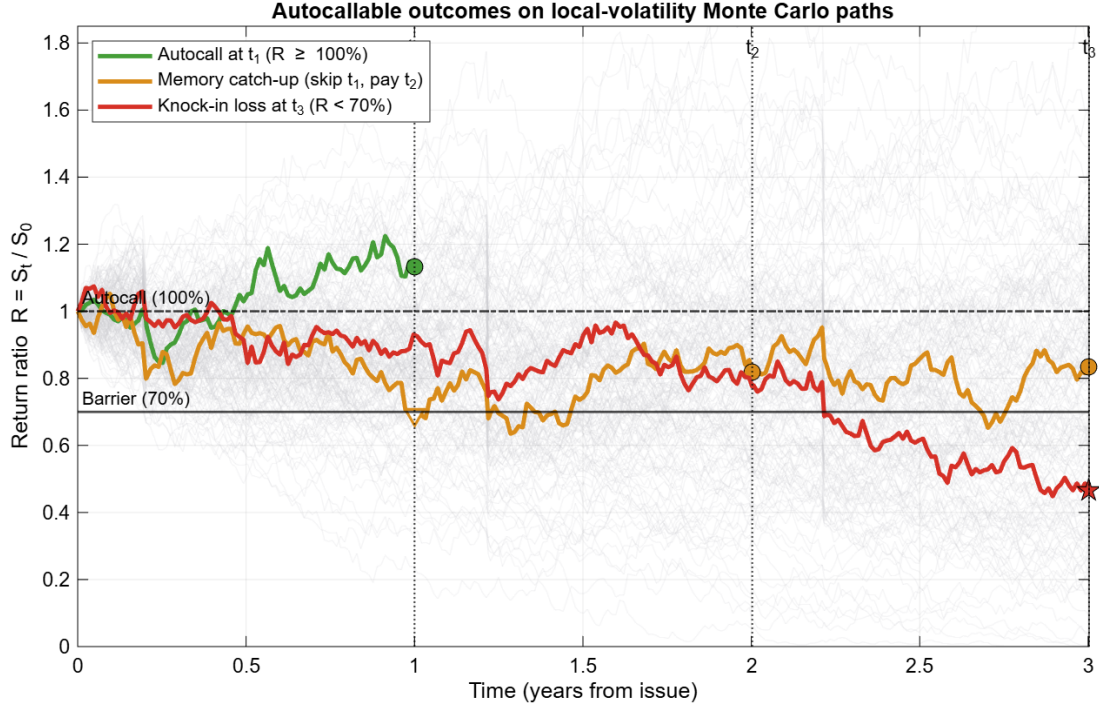


Figure 5: Three representative Monte Carlo trajectories: a path autocalled at t_1 (redeemed early at par plus coupon), a path that skips a coupon and recovers it via the memory snowball, and a path that breaches the 70% barrier at maturity and triggers the knock-in loss.

5.2. Monte Carlo pricing engine

With $r = q = 0$ the certificate price is the risk-neutral expectation of its path-dependent payoff, estimated by Monte Carlo. Paths are generated with a log-Euler scheme; for the local volatility model the step reads

$$S_{k+1} = S_k \exp\left(\sigma_{LV}(S_k) \sqrt{\Delta t_k} Z_k - \frac{1}{2} \sigma_{LV}(S_k)^2 \Delta t_k\right), \quad Z_k \sim \mathcal{N}(0, 1), \quad (14)$$

where the Itô term $-\frac{1}{2}\sigma^2\Delta t$ keeps the discretized process a martingale, $\mathbb{E}[S_{k+1} | S_k] = S_k$, so no spurious drift is introduced (the forward declines only through the dividend jumps). Because σ_{LV} depends on the current state, the LV scheme runs step by step:

```
sigma_k = max(eval_lv(S(:,k), logK_nodes, sigma_nodes), 1e-4);
S(:,k+1) = S(:,k) .* exp( sigma_k.*sqrt(dt(k)).*Zk - 0.5*sigma_k.^2*dt(k) );
S(:,k+1) = S(:,k+1) * (1 - d_ratios(i)); % proportional dividend jump at tau_i
```

The Black–Scholes simulator, having a constant volatility, is instead fully vectorized through a `cumsum` of the log-increments. Variance is reduced with antithetic variates and, for the finite-difference Greeks of Section 6, with Common Random Numbers (the same draws Z shared between base and bumped runs).

Non-uniform time grid. A subtle but important point concerns *where* on the time axis the contractual events fall. The autocall/coupon observations t_1, t_2, t_3 and the ex-dividend times τ_i generally do not coincide with the nodes of a uniform partition. Rather than reading the spot at the nearest node — a snap of up to half a step (a couple of days for $N_t = 250$) that biases both the barrier/autocall test and the timing of the dividend jump — we anchor the exact dates onto the grid:

```
t_grid = sort(unique([linspace(0, T, Nt+1), tau_divs(:)', t_extra(:)']));
```

The union with `tau_divs` and `t_extra` (the observation dates) guarantees that every dividend detaches and every digital test is evaluated at its exact contractual instant. This is the Monte Carlo counterpart of the non-uniform PDE grid of §3.5: it removes the $\mathcal{O}(\Delta t)$ timing bias at no meaningful cost, which matters because the payoff is digital at the 70% barrier and the autocall trigger.

5.3. Par coupon

The fair (par) coupon X ($X\%$) is the annual rate that makes the risk-neutral expected payoff equal the notional. With no discounting,

$$\mathbb{E}^{\mathbb{Q}}[\Pi(X)] = \mathcal{N}, \quad (15)$$

which we solve with a one-dimensional root finder (`fzero`) on the Monte Carlo payoff $\Pi(X)$, monotone increasing in X .

5.4. Vanilla cross-check: validating the engine

Before pricing the structured note we validate the simulator on European puts (ATM-forward and 80% strike, maturities 1/2/3Y). Each option is priced three ways: under the local volatility model (LV-MC), under Black–Scholes by Monte Carlo with the market implied volatility matched to that option’s own strike and maturity (BS-MC), and with the closed-form Black–Scholes formula at the same volatility (BS-an). Table 3 reports the results on the trade date.

Put	Model	1Y	2Y	3Y
ATMF	LV-MC	9.21%	12.34%	14.27%
	BS-MC	9.57%	12.70%	14.56%
	BS-an	9.59%	12.73%	14.57%
80%	LV-MC	4.21%	10.34%	16.48%
	BS-MC	4.48%	10.69%	16.73%
	BS-an	4.52%	10.72%	16.77%

Table 3: European put prices (% of spot). Statistical errors are at the second decimal.

Because the path-dependent autocallable certificate lacks an analytical closed-form pricing solution, we validate our Monte Carlo engine on European put options where exact closed-form Black–Scholes prices (*BS-an*) are available. This cross-check serves as our primary engine validation. First, *BS-MC* matches *BS-an* within Monte Carlo error (e.g., 14.56% vs 14.57% for the 3Y ATMF put), verifying that the path generator and proportional-dividend jumps are correctly implemented. Second, *LV-MC* sits systematically 0.2–0.4 percentage points below the matched *BS* benchmark. This difference is driven by the volatility skew, which steepens from $\approx 24\%$ ATM to $\approx 32\%$ at the 70% strike. Any flat-volatility model ignoring this smile will significantly misprice the note, whose risk is concentrated at the barrier.

5.5. Autocallable price

We now solve (15) for the certificate under both models. The Black–Scholes benchmark is the market-standard single-volatility approach: a flat volatility anchored at the money (the ATM implied volatility 25.5% for the coupon solve; the 3Y ATM-forward implied volatility for the Monte Carlo price). Table 4 gives the par coupons; Table 5 gives the certificate price when *both* models are quoted at the common LV par coupon, so that the two prices differ only through the volatility model.

Pricing Model	Fair Coupon (p.a.)	Price 95% CI [EUR]
Local Volatility (LV)	14.24%	[99,767, 100,233]
Black–Scholes (flat ATM)	15.34%	[99,782, 100,218]

Table 4: Par coupon under the calibrated LV surface and the flat ATM BS benchmark.

5.6. Smile drag and model risk

At the common LV par coupon the flat benchmark prices the certificate 3.9% of notional below its LV value (96.29% vs 100.16%); equivalently, the flat model must offer roughly 110 bp more coupon to reach par (15.34%

Model	Autocallable Price (% Notional)	95% CI
LV-MC	100.16%	[99.92, 100.39]
BS-MC	96.29%	[96.07, 96.51]

Table 5: Autocallable price under LV and the flat Black–Scholes benchmark.

vs 14.24%). The gap is not simulation noise — the engine cross-check is unbiased — and it does not come from a mismatched volatility level. It comes from a feature of the skew that a flat model cannot see.

The note is, in effect, a portfolio of digitals. The dominant economic driver of the certificate is the memory coupon: it pays $X\mathcal{N}$ at every observation date for which $R \geq 70\%$, with snowball recovery of any missed payments. This is a *cash-or-nothing digital* struck at 70%, replicated at the three annual observations; the autocall is a second digital struck at 100% at t_1 and t_2 . The certificate’s price is therefore primarily a linear functional of the risk-neutral barrier probabilities $\mathbb{Q}(R \geq 70\%)$ and $\mathbb{Q}(R \geq 100\%)$ at the observation dates.

Digitals live on the slope of the smile. Replicating a digital call by an infinitesimal call spread and accounting for the strike-dependent volatility yields the standard skew correction

$$\text{digital call}(K) = \text{digital}_{\text{BS}}(K) - \text{Vega} \cdot \sigma'(K). \quad (16)$$

With ISP’s downward equity skew $\sigma'(K) < 0$ the digital is worth strictly *more* than its flat-volatility value, and the correction is largest where the skew is steepest — precisely at the 70% coupon barrier. Each unit of coupon is therefore worth more under the skew-consistent surface than under the flat benchmark, and the autocall digital points the same way.

The terminal density makes it concrete. Figure 6 shows the simulated risk-neutral law of $R = S_{T_3}/S_0$ under both models, matched to the same forward. The downward skew thins out a deep crash tail far below 70% and lifts the bulk upward, so under LV the survival probability at the coupon barrier is materially higher: $\mathbb{Q}_{\text{LV}}(R \geq 70\%) = 54.2\%$ versus $\mathbb{Q}_{\text{BS}}(R \geq 70\%) = 47.0\%$, split as 31.4% vs 26.1% in the coupon-only band and 22.9% vs 20.9% above the autocall level. The flat model compensates with a fatter *ordinary* left tail (52.9% vs 45.8% below 70%), but those paths do not enter the digital payoffs — they only deepen the embedded down-and-in put, which is a second-order offset.

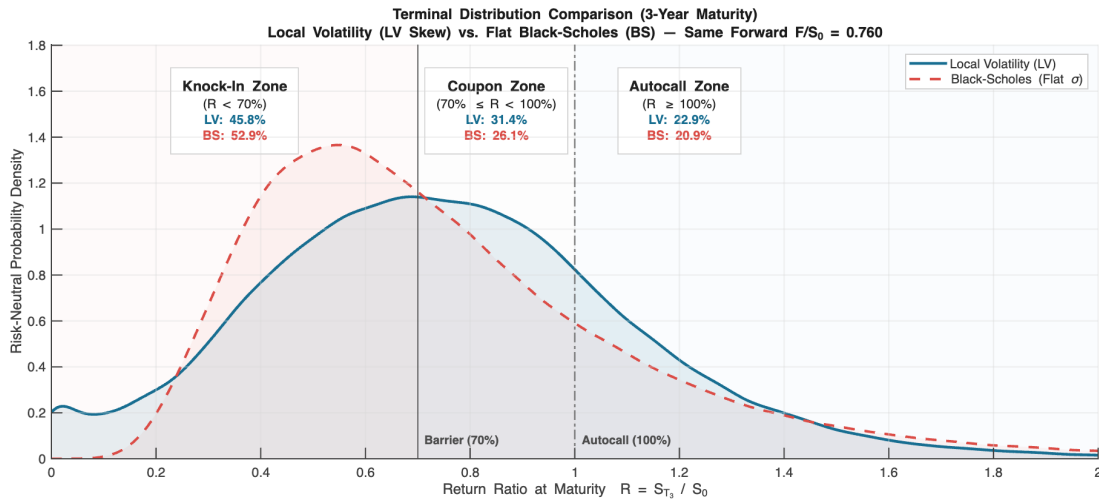


Figure 6: Risk-neutral terminal distribution of $R = S_{T_3}/S_0$ under the calibrated local-volatility surface (solid) versus a flat Black–Scholes volatility (dashed, $\sigma = 3Y$ ATMF IV), both with identical proportional dividends and therefore the *same* forward. The skew lifts the survival probability at the 70% coupon barrier and at the 100% autocall level, which is what the certificate is paid on.

Model risk. The 3.9% / 110 bp gap is therefore pure *model risk*: the price of collapsing the entire volatility skew onto a single number for a payoff whose value lives at the 70% and 100% digital levels. A desk pricing the structure off a flat volatility would systematically over-quote the coupon and overpay the investor — a material

mispricing on the EUR 20M book — which is why the smile-consistent local volatility surface is adopted as the production pricing model and underpins the vega-bucket and dividend-risk analysis that follows.

5.7. Dividend Risk Amplification: Autocallable vs. Vanilla

To highlight the impact of dividend risk, we compare the dividend sensitivity of the autocallable certificate against a standard European vanilla Put option. For a vanilla Put with strike K and maturity T , higher expected dividends lower the forward price ($F_t \downarrow$), monotonically increasing the Put option value. The dividend sensitivity is smooth and well-defined:

$$\text{DivSens}_{\text{van}} = \frac{\partial V_{\text{van}}}{\partial D_k}. \quad (17)$$

Since the option's spot Delta is bounded, this sensitivity remains well-behaved across all spot levels.

In contrast, the path-dependent features of the autocallable certificate amplify this sensitivity through two main channels:

1. **Autocall Obstruction (Duration Extension):** Higher dividends depress the forward price path, reducing the probability of early redemption at the 100% autocall level. This extends the expected life (duration) of the certificate, forcing the desk to remain short the embedded options and short volatility for a longer period. This maturity extension risk generates a non-linear increase in the net dividend risk exposure.
2. **Barrier Breach Discontinuity (Digital Leverage):** The certificate features a maturity-only down-and-in barrier at 70%. If breached, it triggers a discontinuous 30% drop in principal redemption. A downward shift in the terminal stock distribution driven by higher dividends dramatically increases the probability of crossing this barrier. This digital-like cliff produces a severe localized spike in dividend sensitivity (Dividend Gamma) near the barrier that is orders of magnitude larger than the smooth sensitivity of a vanilla Put.

6. Greeks Cross-Validation and Dividend Dynamics

To calculate the risk sensitivities (Greeks) near the highly non-linear down-and-in barrier (70% moneyness) and autocall observation levels, standard independent Monte Carlo estimators suffer from excessive sampling noise. To achieve precise, stable gradients for delta and vega rebalancing, we implement two variance reduction techniques: **Common Random Numbers (CRN)** and **Risk Profile Pinning**.

6.1. Modular Computation Tree and Runtime Layerization

To achieve maximum modularity, numerical alignment, and computational speed, the risk management engine is structured as a decoupled runtime computation tree. This modular architecture is designed specifically to integrate the Common Random Generators (CRG) / Common Random Numbers (CRN) framework, ensuring that a single set of synchronized paths is shared across all portfolios, as seen in the chart below (Figure 7).

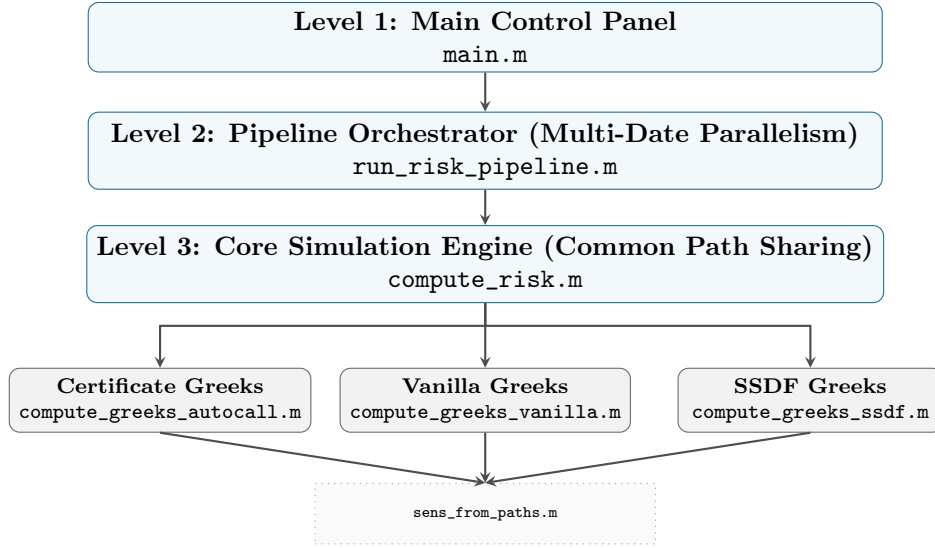


Figure 7: Runtime Greeks computation layering. High-level orchestrators govern execution flow and share path simulations under Common Random Numbers, dispatching to specialized calculators and low-level finite-difference gradient estimators.

This five-level layered tree decouples workflow orchestration, simulation path generation, contract payoff evaluation, and mathematical differentiation. This decoupling ensures that paths are simulated once per date and shared across all instruments, significantly reducing CPU runtime while guaranteeing full numeric consistency for joint delta-vega hedging.

6.2. Variance Reduction via Common Random Numbers (CRN)

To compute finite-difference sensitivities (Delta, Gamma, Vega, and Dividend Sensitivity), we reject independent path simulations in favor of the Common Random Numbers (CRN) technique. Let θ represent the Greek of interest, and $\hat{V}(\mathbf{Z}; \epsilon)$ be the Monte Carlo price estimator of a contract evaluated under a parameter shift ϵ , driven by a matrix of independent standard normal random variables $\mathbf{Z}$. For compactness, we denote $\hat{V}^\pm \equiv \hat{V}(\mathbf{Z}; \pm\epsilon)$. Under the central difference scheme, the estimator $\hat{\theta}$ is formulated as:

$$\hat{\theta}(\mathbf{Z}) = \frac{\hat{V}^+ - \hat{V}^-}{2\epsilon} \quad (18)$$

where the *exact same* random matrix $\mathbf{Z}$ is shared between the base, upward, and downward perturbed simulations. The statistical variance of this sensitivity estimator evaluates to:

$$\text{Var}(\hat{\theta}(\mathbf{Z})) = \frac{1}{4\epsilon^2} \left[\text{Var}(\hat{V}^+) + \text{Var}(\hat{V}^-) - 2 \text{Cov}(\hat{V}^+, \hat{V}^-) \right] \quad (19)$$

By generating the underlying asset paths utilizing identical Brownian increments $\mathbf{Z}$ across the baseline and bumped parameter spaces, the prices $\hat{V}^+$ and $\hat{V}^-$ become highly positively correlated, yielding $\text{Cov}(\hat{V}^+, \hat{V}^-) \approx$

$\text{Var}(\hat{V}^\pm)$. This correlation term cancels out the majority of the background statistical noise in (19), reducing the standard error of the Greek estimator by several orders of magnitude and isolating the pure mathematical sensitivity to the shocked parameter.

6.3. Implementation of Common Random Numbers in the Pricing Engine

Our modularized code enforces CRN at the highest level of the execution pipeline. In the risk management runner, the random increment matrix is pre-generated once per valuation date on the global simulation time-grid, cached inside the workspace, and explicitly broadcast to the parallel cores.

In the runner file `src/orchestrators/run_risk_pipeline.m`, the random normal matrix Z is defined for the first time on line 37:

```
T_i = ts_i.T3;
tg_i = sort(unique([linspace(0, T_i, NT_MC+1), mkt.div_dates(:)', T_obs(i,:)]')));
Z = randn(N_PATHS_GREEK, numel(tg_i)-1);
Z_all{i} = Z;
```

The pre-generated matrix Z is subsequently passed to the core risk calculator `src/orchestrators/compute_risk.m`, which simulates the entire set of baseline and perturbed trajectories using the exact same random sequence:

```
% (1) Simulate baseline paths
S_base = simulate_lv_paths_MC(mkt.spot, mkt.nodes_K, mkt.sigma_nodes, ...
    T, Nt, mkt.div_dates, mkt.d, N_paths, false, Z, t_extra);

% (2) Spot bumped paths for Delta & Gamma
S_up = simulate_lv_paths_MC(spot_up, K_up, sig_up, T, Nt, ...
    mkt.div_dates, d_up_s, N_paths, false, Z, t_extra);
S_down = simulate_lv_paths_MC(spot_down, K_down, sig_down, T, Nt, ...
    mkt.div_dates, d_down_s, N_paths, false, Z, t_extra);

% (3) Volatility bumped paths for Vega Bucket (CRN Recalibration)
S_sup = simulate_lv_paths_MC(mkt.spot, K_sup, sig_sup, T, Nt, ...
    mkt.div_dates, mkt.d, N_paths, false, Z, t_extra);
```

Because all candidate instruments (the short certificate, vanilla ATM options, and SSDF hedging contracts) are priced on these highly correlated, matched path sets, their computed sensitivities are mathematically consistent. This guarantees that our vega-gamma profile matching is entirely free of sampling asymmetric noise, protecting the desk from spurious trading signals.

6.4. Noise-Free Profile Pinning at the Inception Spot

When projecting the Vega and Dividend Sensitivity profiles across the spot price grid ($0.4S_0 \leq S \leq 1.2S_0$), the grid is constructed utilizing a non-homogeneous node spacing to ensure that the current spot price S_0 is represented exactly, without requiring interpolation.

However, evaluating sensitivities across a grid of independent spot points introduces small localized sampling variations due to the independent paths simulated at each grid node. To eliminate this Monte Carlo grid noise at the highly critical current market spot, we implement a *Profile Pinning* strategy.

Let $\theta(S^{(k)})$ represent the raw Monte Carlo profile of a Greek computed across the spot grid nodes $S^{(k)}$. We identify the exact index k_0 corresponding to the trade date spot $S^{(k_0)} \equiv S_0$. We then overwrite the grid estimator at this node with the high-precision baseline sensitivity $\theta(S_0)$ computed at t_0 using the full simulation path:

$$\theta(S^{(k_0)}) = \theta(S_0) \quad (20)$$

In our visual risk profile pipeline `src/orchestrators/run_risk_profile_pipeline.m`, this pinning is implemented as follows:

```
% Pin profiles at S0 to exact t0 greeks to remove Monte Carlo grid noise
[~, idx_s0] = min(abs(spot_grid - mkt{1}.spot));
profile_cert.vega(idx_s0) = greeks_t0.cert.vega;
profile_cert.div_sens(:, idx_s0) = greeks_t0.cert.div_sens;
profile_van.vega(idx_s0) = greeks_t0.vanilla.vega;
profile_van.div_sens(:, idx_s0) = greeks_t0.vanilla.div_sens;
profile_van_opt.vega(idx_s0) = vanilla_opt.greeks.vega;
profile_van_opt.div_sens(:, idx_s0) = vanilla_opt.greeks.div_sens;
```


This pinning procedure guarantees that at the current equity spot level (where the desk maintains its active portfolio), the hedging ratios remain mathematically exact and perfectly consistent with the historical pricing book, removing any risk of trading execution errors driven by simulation artifacts.

6.5. Analytical Proof of Zero SSDF Delta and Vega by Construction

In our discrete proportional dividend framework, the spot price jumps on the ex-dividend date τ_i according to:

$$S(\tau_i^+) = S(\tau_i^-)(1 - d_i), \quad (21)$$

where d_i is the proportional dividend yield. The model price of the i -th Single Stock Dividend Future (SSDF) contract, which pays the cash dividend detaching at τ_i under zero interest, funding, and repo rates, is given by the risk-neutral expectation:

$$SSDF_i^{model}(S_0) = \mathbb{E}^{\mathbb{Q}}[D_i] = \mathbb{E}^{\mathbb{Q}}[d_i S(\tau_i^-)] = d_i S_0 \prod_{j=1}^{i-1} (1 - d_j). \quad (22)$$

Since the SSDF is a liquid, market-traded instrument, its model price must be matched to the observed market quote $SSDF_i^{mkt}$ to prevent arbitrage:

$$SSDF_i^{model}(S_0) \equiv SSDF_i^{mkt}. \quad (23)$$

When computing the finite-difference Delta of any instrument, we perturb the inception spot $S_0 \rightarrow S_0 \pm h$. To maintain calibration to the market, the proportional dividend yields $\{d_i\}$ cannot be treated as constant parameters; instead, they must be dynamically re-bootstrapped from the fixed market quotes $\{SSDF_k^{mkt}\}$:

$$d_i(S_0 \pm h) = \frac{SSDF_i^{mkt}}{(S_0 \pm h) \prod_{j=1}^{i-1} (1 - d_j(S_0 \pm h))}. \quad (24)$$

Equation (24) defines the spot-dependent adjustment of the proportional dividend yields required to match market quotes. By substituting this relationship back into the model price equation (22), we obtain:

$$SSDF_i^{model}(S_0 \pm h, \{d_j(S_0 \pm h)\}) = d_i(S_0 \pm h) \cdot (S_0 \pm h) \prod_{j=1}^{i-1} (1 - d_j(S_0 \pm h)) \equiv SSDF_i^{mkt}. \quad (25)$$

Because the model-calibrated SSDF price is invariant under perturbations of S_0 , its finite-difference Delta is identically zero:

$$\Delta_{SSDF_i} = \frac{SSDF_i^{model}(S_0 + h) - SSDF_i^{model}(S_0 - h)}{2h} = \frac{SSDF_i^{mkt} - SSDF_i^{mkt}}{2h} \equiv 0. \quad (26)$$

Similarly, the market quote $SSDF_i^{mkt}$ is an input parameter that is completely independent of the implied volatility surface. Bumping the volatility surface leaves the model-calibrated SSDF price unchanged, meaning its Vega is also exactly zero:

$$\mathcal{V}_{SSDF_i} = \frac{SSDF_i^{model}(\sigma + \epsilon) - SSDF_i^{model}(\sigma - \epsilon)}{2\epsilon} \equiv 0. \quad (27)$$

This proof demonstrates that the SSDF contracts are pure, orthogonal instruments for hedging dividend risk. Because they have zero Delta and zero Vega, they allow the desk to isolate dividend risk from spot directional risk and volatility risk, establishing a decoupled risk management framework.

7. Hedge Portfolio Design & Construction

7.1. “What’s Going On?” – Book Position & Risk Diagnostics

To establish a clear risk management framework, we begin by analyzing the client-facing transaction that has landed in the trading desk’s book. The desk has sold a total notional of **EUR 20 million**, corresponding to $N_{\text{CERT}} = -200$ structured contracts (each with a face value of EUR 100,000) written on Intesa Sanpaolo (ISP). This structured position exposes the book to a highly non-linear, multi-dimensional risk regime that must be thoroughly understood and systematically immunized.

Figure 8 displays the standalone risk fingerprints (Vega and dividend sensitivities) of this short certificate position in isolation.

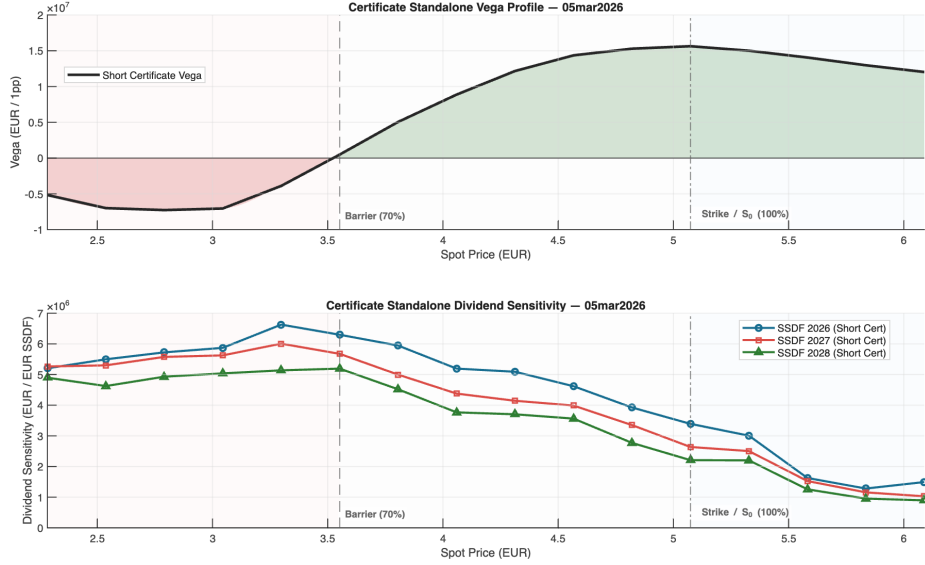


Figure 8: Standalone risk diagnostics for the short EUR 20M certificate book.

The standalone profiles reveal the following critical risk dynamics:

7.1.1 Volatility Risk Profile (Vega)

The top panel of Figure 8 illustrates the standalone Vega of our short position (measured in EUR per 1 percentage point shift in implied volatility). Because the certificate has a short-volatility profile for the holder (i.e., higher volatility increases the likelihood of a down-and-in barrier breach and coupon loss), our short book position has a **positive Vega** of approximately **EUR 15.6 million / 1pp** at the inception spot $S_0 = 5.074$ EUR (100% of the strike). As the spot price drifts above S_0 , the Vega remains positive but gradually decays as the autocall feature becomes highly likely to trigger, dampening volatility sensitivity. Conversely, as the spot falls toward the barrier the Vega remains high and then **flips sign below it** ($S < 3.55$ EUR, red region): in the knocked-in regime the residual optionality reverses sign, so the short book gains from volatility spikes. This non-monotone profile already exposes the structural inadequacy of a flat-volatility hedge — any static position calibrated at S_0 suffers severe **Vega leakage** as the spot drifts away.

7.1.2 Dividend Risk Profile (Dividend Sensitivity)

The bottom panel of Figure 8 displays the dividend sensitivity of the short certificate position across the three tenors. Because higher cash dividends depress the forward stock price and reduce the certificate’s value for the holder, our short book position is **short dividends** (possessing positive sensitivity, up to **EUR 6.5 million / EUR SSDF** for the 2026 tenor). These sensitivities are also highly non-linear and spot-dependent, peaking just below the barrier (around $S = 3.25$ EUR) and falling off as the spot moves far above S_0 . This spot-dependence indicates that a simple dividend hedge matched only at S_0 will inevitably leave residual dividend exposures when the spot moves.

To immunize these complex, non-linear risk exposures, the desk designs and evaluates four static hedging portfolios of increasing complexity, which are constructed at inception (t_0):

- **Portfolio A (Delta-Only)**: Neutralizes spot directional risk using the underlying stock.
- **Portfolio B (Delta + Dividend)**: Adds Synthetic Stock Dividend Futures (SSDF) to neutralize first-order dividend sensitivities.
- **Portfolio C (Delta + Dividend + ATMF Put)**: Adds a 3Y At-The-Money-Forward Put to neutralize parallel Vega at S_0 .
- **Portfolio D (Delta + Dividend + Optimal Put)**: Replaces the ATMF Put with an optimized Put selected via global L2 minimization of Vega and dividend residuals over the spot grid.

7.2. Hedge Portfolio Design and Mathematical Formulation

The static hedges are constructed at inception (t_0) using the following instruments:

- **ISP Common Stock**: Used to neutralize spot directional risk (Delta).
- **Synthetic Stock Dividend Futures (SSDF)**: Three liquid tenors (Maturities: May 2026, May 2027, May 2028) with market prices of 0.3948 EUR, 0.4138 EUR, and 0.4109 EUR, used to hedge discrete proportional dividend risks.
- **Vanilla Put Options**: Used to neutralize the book's parallel Vega exposure. Portfolio C uses a standard **3Y At-The-Money-Forward Put** to hedge parallel volatility shifts. Portfolio D uses an **optimal 110% Put** selected by minimizing the probability-weighted L2 residual of the hedged portfolio over the spot grid. For each candidate vanilla, the static Vega and SSDF hedge is rebuilt at S_0 , and the residual Vega and dividend sensitivity are evaluated across the grid. The selected put is therefore chosen not only to cancel Vega at inception, but to better match the certificate's risk profile away from S_0 .

The mathematical formulation for the net Greeks of the hedged book is:

$$\text{Net } \Delta_p = N_{\text{CERT}} \cdot \Delta_{\text{cert}} + h_{\text{spot}} \cdot S_0 + \sum_{k=1}^3 h_{\text{SSDF},k} \cdot \Delta_{\text{SSDF},k} + h_{\text{van}} \cdot \Delta_{\text{van}} = 0 \quad (28)$$

$$\text{Net DivSens}_{p,k} = N_{\text{CERT}} \cdot \text{DivSens}_{\text{cert},k} + h_{\text{SSDF},k} + h_{\text{van}} \cdot \text{DivSens}_{\text{van},k} = 0 \quad \forall k \in \{1, 2, 3\} \quad (29)$$

$$\text{Net } \mathcal{V}_p = N_{\text{CERT}} \cdot \mathcal{V}_{\text{cert}} + h_{\text{van}} \cdot \mathcal{V}_{\text{van}} = 0 \quad (30)$$

7.3. Progression of Hedges

1. **Portfolio A (Delta-Only)**: The desk replicates spot risk using a simple stock-only position ($h_{\text{spot}} = +2,846,606$ shares, cost = 14,443,681 EUR). By ignoring dividend and volatility risks, this portfolio is highly vulnerable to dividend yield leakages and volatility shocks.
2. **Portfolio B (Delta + Dividend)**: The desk introduces SSDF contracts to neutralize the first-order dividend sensitivity ($h_{\text{SSDF}} = [-1.34, -1.09, -0.91]$ MEUR, total SSDF cost = -3,336,383 EUR). Proportional dividend risk is isolated, but the book remains exposed to massive Vega shocks, particularly near the down-and-in barrier and the autocall levels.
3. **Portfolio C (Delta + Dividend + ATMF Vega)**: The desk adds a standard 3Y ATMF Put option ($h_{\text{van}} = -6,078,691$ contracts, cost = -4,413,728 EUR) to hedge parallel volatility shifts. While parallel Vega is neutralized at the current spot S_0 , this portfolio suffers from severe "Vega leakage" as the spot price moves away from S_0 due to the rapid decay of the ATMF option's Vega.
4. **Portfolio D (Delta + Dividend + Optimal Vega)**: The ATMF Put of Portfolio C is replaced by the optimized Put selected over the candidate grid of strikes and maturities. For each candidate vanilla v , the hedge is first constructed at S_0 by cancelling parallel Vega and dividend sensitivity, and the resulting residual risk is then measured across the full spot grid.

The selected vanilla minimizes the probability-weighted L2 residual

$$L_{\text{tot}}(v) = \underbrace{\sqrt{\sum_k w_k \left(R_{\text{Vega}}^v(S_k) \right)^2}}_{L_{\text{Vega}}(v)} + \underbrace{\sqrt{\sum_k w_k \sum_{j=1}^3 \left(R_{\text{Div},j}^v(S_k) \right)^2}}_{L_{\text{Div}}(v)}.$$

Here $R_{\text{Vega}}^v(S_k)$ is the residual net Vega of the hedged portfolio at spot node S_k , and $R_{\text{Div},j}^v(S_k)$ is the residual dividend sensitivity with respect to the j -th SSDF tenor.

The weights w_k are obtained from a risk-neutral lognormal density over the spot grid:

$$w_k = \frac{f_{\text{LN}}(S_k) \Delta S_k}{\sum_{\ell} f_{\text{LN}}(S_{\ell}) \Delta S_{\ell}},$$

with

$$f_{\text{LN}}(S) = \frac{1}{S \sigma \sqrt{\tau} \sqrt{2\pi}} \exp \left(-\frac{(\log(S/S_0) - \mu)^2}{2\sigma^2 \tau} \right), \quad \mu = \log \left(\frac{F_\tau}{S_0} \right) - \frac{1}{2} \sigma^2 \tau.$$

This weighting gives more importance to spot scenarios that are more likely under the risk-neutral transition density.

In the final hedge, Portfolio D uses $h_{\text{van}} = -5.91$ M contracts of the optimal 110% Put, together with adjusted SSDF and stock positions. Like Portfolio C, it is exactly neutral in Delta, dividend sensitivity and parallel Vega at S_0 , but the chosen vanilla is designed to reduce residual risk away from the current spot.

7.4. Portfolio Hedges and Ratios Table (t0 = 05-Mar-2026)

Table 6 details the static hedge positions constructed at inception ($S_0 = 5.0740$ EUR).

Instrument / Leg	Portfolio A	Portfolio B	Portfolio C	Portfolio D
ISP shares	+2,846,606	+2,846,572	+713,809	+249,117
ISP cost (EUR)	+14,443,681	+14,443,509	+3,621,865	+1,264,020
SSDF May 2026 (EUR)	—	−1,339,514	−258,297	−94,180
SSDF May 2027 (EUR)	—	−1,091,774	−28,883	+148,774
SSDF May 2028 (EUR)	—	−905,095	+55,395	+242,123
SSDF total (EUR)	—	−3,336,383	−231,785	+296,717
Vanilla Put (contracts)	—	—	−6,078,691	−5,914,689
Vanilla Put cost (EUR)	—	—	−4,413,728	−5,588,954
TOTAL COST (EUR)	+14,443,681	+11,107,126	−1,023,648	−4,028,217
% of Book Notional	72.22%	55.54%	−5.12%	−20.14%

Table 6: Static Hedge Portfolios for Short EUR 20M Certificate Position ($N_{\text{CERT}} = -200$)

8. Standing Risk Profiles & Hedging Efficiency

Having analyzed the standalone risk profile of the short certificate position in Section 7.1 (and illustrated in Figure 8), we now evaluate the net Greeks of the hedged book across the spot grid to measure the hedging efficiency of each portfolio.

8.1. Standing Greeks vs. Spot Grid

8.1.1 Net Vega Profile & Vega-Gamma Profile Matching

Aspect	Observation
Quantitative finding	At inception spot S_0 , both Portfolios C and D maintain a net Vega of exactly zero by construction. Once the spot moves away from S_0 , however, Portfolio C develops a larger residual Vega because the ATMF put profile no longer offsets the certificate profile well across the grid; Portfolio D keeps the net Vega closer to zero in the probability-weighted region targeted by the L2 optimization, since the selected 110% put provides a better profile match.
Financial intuition	The certificate's short position is embedded with a short European down-and-in put. Near the barrier ($0.7S_0$) the certificate's Vega surges because any downward spot move increases the probability of breaching it and triggering the principal knock-in loss. The ATMF Put in Portfolio C peaks at S_0 ; as the spot falls it moves deep in-the-money, losing its optionality and hence its Vega, leaving the desk exposed near the barrier. Portfolio D addresses this mismatch by selecting a strike whose Vega profile better tracks the certificate's.
Hedging implication	Portfolio C is a textbook hedge that fails under stress scenarios where the spot drifts toward the barrier. Portfolio D is more robust because the optimized put makes the hedge less sensitive to moderate spot moves and limits the Vega leakage.

8.1.2 Net Dividend Sensitivity Profile & Hedging Mismatch

Aspect	Observation
Quantitative finding	The net dividend sensitivity of Portfolios B and C remains close to zero across the spot grid but shows small residual variations as the spot moves away from S_0 . Portfolio D keeps the dividend-sensitivity residuals more concentrated around zero in the relevant spot region, because the optimal vanilla is selected jointly on Vega and dividend-sensitivity residuals.
Financial intuition	Under the bootstrapped proportional dividend framework, the expected cash dividends are locked to the fixed SSDF market prices, ensuring that the SSDF has zero Delta by construction. The certificate's dividend sensitivity is however non-linear in spot (due to the optionality of the barrier and autocall), whereas the SSDF dividend sensitivity is constant and linear: a static hedge matching sensitivities at S_0 leaves small, non-linear residuals as the spot moves away.
Hedging implication	The residual dividend sensitivity is small in practice, confirming that the SSDF-based hedge successfully isolates dividend risk from equity spot risk across realistic spot moves.

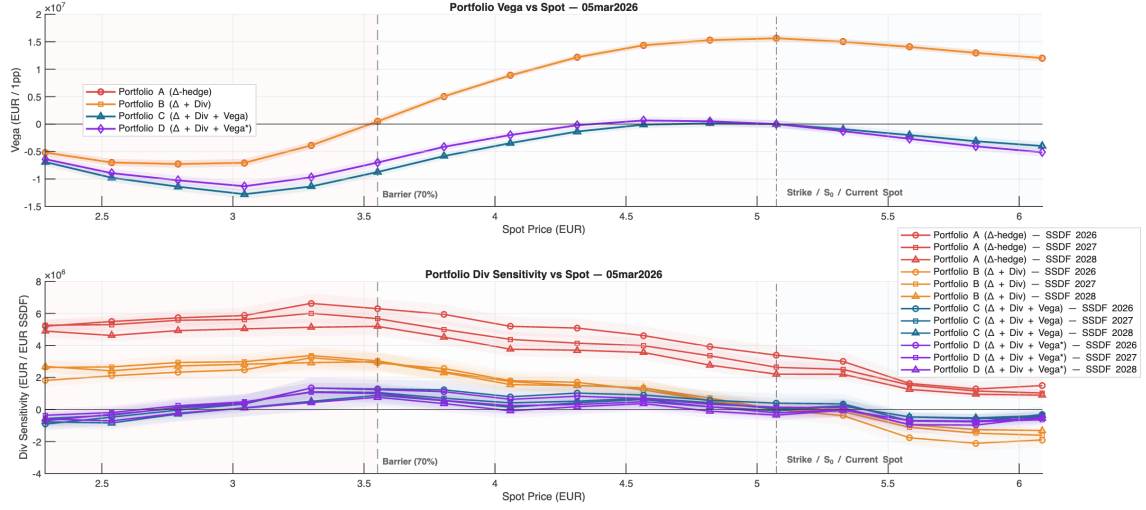


Figure 9: Risk profiles of the hedged book across the four portfolios. Top: Net Vega profiles (EUR/1pp) across the spot price grid (displaying barrier at $0.7S_0$ and strike/autocall at S_0). Bottom: Net dividend sensitivity profiles (EUR/EUR SSDF) across the spot grid for the three tenors (2026, 2027, 2028).

9. P&L Explain & Model Attributions

9.1. Validation under Next Days Market Shocks

We evaluate the performance of the four portfolios across two realistic stress scenarios drawn from live ISP market data at future dates: t_1 (06-Mar-2026) and t_2 (09-Mar-2026). Table 7 reports the observed moves in spot and SSDF prices across the three tenors.

	t_0 (05-Mar-2026)	t_1 (06-Mar-2026)	t_2 (09-Mar-2026)
Instrument	Price	Price Δ	Price Δ
ISP Spot (EUR)	5.0740	5.2770 +4.00%	4.2216 -16.80%
SSDF May 2026 (EUR)	0.3950	0.4060 +2.78%	0.1020 -74.18%
SSDF May 2027 (EUR)	0.4140	0.4250 +2.66%	0.1700 -58.94%
SSDF May 2028 (EUR)	0.4100	0.4220 +2.93%	0.2110 -48.54%

Table 7: Market data across the three evaluation dates. The t_2 scenario corresponds to a sharp ISP sell-off (-16.8%) accompanied by a dramatic collapse in SSDF prices (down -49% to -74%), reflecting the market’s reassessment of future dividend expectations.

SSDFs are listed instruments on exchanges, and as such they have an observable market price every day, just like the stock price. The market quotes them freely, and their price reflects the aggregate expectations of participants on ISP’s future dividends. When at t_2 the spot collapses by 16.8%, the market simultaneously revises its expectations on future dividends—and the SSDFs collapse by up to 74%. This move is not derived from the spot via a formula in our model. Rather, it is an independent market input, just like the fact that the smile did not move between t_0 and t_2 . Therefore, our model is consistent with reality. In the model, the SSDF

is exogenous, implying $\Delta_{SSDF} \equiv 0$. The net P&L for the desk is decomposed leg by leg as:

$$\text{P\&L}_{\text{net}} = \underbrace{N_{\text{CERT}} \cdot (V_{\text{cert}}(t) - V_{\text{cert}}(t_0))}_{\text{certificate leg}} \quad (31)$$

$$+ \underbrace{h_{\text{spot}} \cdot (S(t) - S_0)}_{\text{spot leg}} \quad (32)$$

$$+ \underbrace{\sum_{k=1}^3 h_{\text{SSDF},k} \cdot \left(\frac{\text{SSDF}_k(t)}{\text{SSDF}_k(t_0)} - 1 \right)}_{\text{SSDF legs}} \quad (33)$$

$$+ \underbrace{h_{\text{van}} \cdot (V_{\text{van}}(t) - V_{\text{van}}(t_0))}_{\text{vanilla leg}}. \quad (34)$$

Table 8 presents the P&L breakdown under the next days' stress scenarios.

P&L Component	Portfolio A	Portfolio B	Portfolio C	Portfolio D
Scenario t1 (06-Mar-2026)				
Certificate P&L (EUR)	−422,009	−422,009	−422,009	−422,009
Spot Leg P&L (EUR)	+577,747	+577,740	+144,875	+50,561
SSDF Leg P&L (EUR)	—	−92,802	−6,339	+8,417
Vanilla Leg P&L (EUR)	—	—	+348,926	+425,241
NET P&L (EUR)	+155,738	+62,929	+65,452	+62,209
% of Book Notional	0.78%	0.31%	0.33%	0.31%
Scenario t2 (09-Mar-2026)				
Certificate P&L (EUR)	+249,240	+249,240	+249,240	+249,240
Spot Leg P&L (EUR)	−2,426,538	−2,426,509	−608,473	−212,355
SSDF Leg P&L (EUR)	—	+2,076,377	+181,733	−135,341
Vanilla Leg P&L (EUR)	—	—	+10,300	−79,676
NET P&L (EUR)	−2,177,298	−100,892	−167,199	−178,132
% of Book Notional	−10.89%	−0.50%	−0.84%	−0.89%

Table 8: P&L Explain and Component Attribution across Evaluation Dates

9.2. Analysis of P&L Residuals & Model Risk

- **Portfolio A (spot-only):**
 - **Spot and Volatility Risk:** Without dividend or vega hedges, the book is fully exposed to volatility moves and spot fluctuations. The large P&L swing at t_2 confirms that delta-only replication is inadequate for structured products with barrier and autocall features.
 - **Dividend Risk Impact:** Portfolio A carries no dividend hedge. When expected dividends rise at t_1 (+2.8%), the forward price path falls, giving the short certificate book an unhedged mark-to-market gain. At t_2 , the −74% dividend collapse pushes the forward path higher, generating an unhedged mark-to-market loss that is masked by the massive spot leg drop.
- **Portfolio B (spot + SSDF):**
 - **Spot and Volatility Risk:** The book carries no vega hedge. Near the barrier at t_2 , the certificate's Vega rises sharply, leaving the desk fully exposed to volatility shocks and skew deformations.
 - **Dividend Risk Impact:** Although the SSDF hedge cancels dividend risk at S_0 , it fails away from it because SSDFs are linear while the certificate is highly non-linear. When spot and dividends collapse at t_2 , the short SSDF positions generate a massive profit of EUR +2,076,377. This gain is economically artificial: the certificate's dividend sensitivity has decayed near the barrier, whereas the SSDF contracts continue to produce linear gains, creating a spurious profit that accidentally masks the unhedged Vega losses.

- **Portfolio C (spot + SSDF + ATMF Put):**
 - **Spot and Volatility Risk:** Adding the ATMF Put neutralizes parallel vega at S_0 , substantially reducing the residual P&L. However, the remaining leakage at t_2 is consistent with the vega mismatch: as the spot drops sharply, the ATMF Put moves deep ITM and loses its Vega, leaving the desk exposed near the barrier.
 - **Dividend Risk Impact:** The ATMF Put option carries its own dividend sensitivity, so the desk sells fewer SSDF contracts at t_0 . Consequently, the dividend collapse at t_2 yields a much smaller SSDF profit of EUR +181,733. However, because the ATMF option moves deep ITM during the sell-off, its dividend sensitivity decays rapidly, exposing the desk to a significant unhedged profile mismatch under stress.
- **Portfolio D (spot + SSDF + Optimal Put):**
 - **Spot and Volatility Risk:** At t_1 , Portfolio D produces a residual P&L of EUR +62,209, which is closer to a perfect hedge than Portfolio C. In the larger t_2 stress scenario, the residual is EUR -178,132. The global L2 optimization minimizes Vega residuals across the entire distribution of likely spot prices, avoiding accidental offsets.
 - **Dividend Risk Impact:** The optimal 110% Put option has a high strike and carries large dividend sensitivities across the grid. To balance this, the desk is net long the later SSDF tenors. When dividends collapse at t_2 , this long position loses EUR -135,341. This loss is economically correct: it is the counterpart of the Put option's gain, which protected the book. The stable net P&L confirms that joint profile matching successfully neutralizes both spot and dividend shocks.

9.3. Sufficiency of the Hedge and Remaining Risks

The P&L evaluation shows that adding dividend and Vega hedges materially improves the stability of the book, but does not fully eliminate residual P&L. Portfolios C and D are neutral in Delta, dividend sensitivity and parallel Vega at inception, but the evaluation across the t_1 and t_2 scenarios still shows residual gains or losses after market moves. This means that the hedges reduce the main first-order exposures at t_0 , but do not fully immunize the book over time.

The remaining unhedged risks are mainly second-order effects:

- **Gamma risk:** the certificate price is highly non-linear in spot near the 70% barrier and the 100% autocall level. A Delta hedge is exact only locally; after a larger spot move, the Delta changes and the hedge is no longer perfect.
- **Vanna risk:** the Vega exposure changes when the spot moves. Therefore, a Vega hedge that is exact at S_0 may become imperfect after the underlying moves, especially near the barrier where the certificate Vega profile changes rapidly.
- **Volga risk:** the price response to volatility shocks is not perfectly linear. The hedge neutralizes a parallel Vega exposure for small volatility bumps, but large volatility moves can still generate residual P&L because the Vega itself changes with volatility.

Portfolio D mitigates part of these risks by selecting the vanilla hedge through a profile-matching criterion rather than only cancelling Vega locally at S_0 . However, it remains a static first-order hedge. A more complete hedge would require additional options across strikes and maturities, together with periodic rebalancing, which is beyond the scope of the static portfolios considered here.

9.4. Desk Profitability & Transaction Cost Analysis (Bid-Offer Spreads)

To evaluate the actual commercial viability of selling the certificate and hedging it, we perform a P&L evaluation incorporating realistic transaction costs (bid-offer spreads). The desk acts as a price-taker in the hedging market (paying half the spread on every hedge leg) and a price-maker in the certificate market (earning half the spread). Table 9 details the hedged P&L explain including bid-offer transaction costs.

P&L Component	Portfolio A	Portfolio B	Portfolio C	Portfolio D
Scenario t1 (06-Mar-2026)				
Gross Net P&L (EUR)	+155,738	+62,929	+65,452	+62,209
One-time Txn Costs (EUR)	+49,298	+47,630	+45,254	+44,126
NET HEDGED P&L (EUR)	+205,036	+110,559	+110,706	+106,335
% of Book Notional	1.03%	0.55%	0.55%	0.53%
Scenario t2 (09-Mar-2026)				
Gross Net P&L (EUR)	−2,177,298	−100,892	−167,199	−178,132
One-time Txn Costs (EUR)	+49,298	+47,630	+45,254	+44,126
NET HEDGED P&L (EUR)	−2,128,000	−53,262	−121,945	−134,007
% of Book Notional	−10.64%	−0.27%	−0.61%	−0.67%

Table 9: Hedged P&L Explain Incorporating Bid-Offer Spreads (Transaction Costs)

- **Desk revenue from spreads:** Since the desk acts as a price-maker for the certificate and a price-taker for the hedging instruments, it earns half of the certificate’s 0.5% spread (corresponding to a credit of +50,000 EUR at inception) while paying spreads on the stock, SSDF, and vanilla options. This structure results in a net positive transaction cost adjustment (revenue) for all portfolios (ranging from +44,126 EUR for Portfolio D to +49,298 EUR for Portfolio A).
- **Positive hedging buffer:** This net spread revenue acts as a positive buffer that directly improves the desk’s net hedged P&L across all scenarios. For example, for the optimized Portfolio D at t_1 , the net P&L increases from a gross of +62,209 EUR to a net hedged profit of +106,335 EUR. Under the severe t_2 sell-off, the net loss is mitigated from −178,132 EUR gross to −134,007 EUR net. This reinforces that the bid-ask spread is a structural source of profitability that significantly cushions the residual hedging error, always supporting the desk in a positive way.
- **Hedging viability:** The fact that the net hedged P&L remains positive under mild scenarios (EUR +106,335 at t_1) and that the transaction cost adjustment is positive in all cases demonstrates the strong commercial viability of the certificate desk. The primary driver of performance remains the quality of the static risk matching (limiting the first-order leakages) rather than transaction cost friction, which is fully offset by the price-making spread capture.

10. Software Architecture

To support calibration, pricing, Greeks calculation, and multi-instrument hedging, the codebase is structured in a modular, layered architecture. The design decouples raw market data processing and numerical grids from contract-specific payoffs and simulation engines.

10.1. Code Organization

The application's entry point is `main.m`, which governs the execution of calibration, pricing, risk management, and portfolio hedging. The script `node_sweep.m` is an auxiliary research file not used in the production pipeline. To streamline the codebase, several utility functions were removed and their outputs formatted and integrated directly within the report printing routines. The pipeline utilizes parallel loops across independent dates and path simulations; to enable this acceleration via `parfor`, installing the MATLAB Parallel Computing Toolbox is recommended.

The implementation is organized into three distinct structural layers:

- **Control Layer** (`src/`): Configures parameters and drives execution.
- **Orchestration Layer** (`src/orchestrators/`): Coordinates high-level data passing and parallel pipeline execution.
- **Functional Layer** (`src/helpers/`): Divided into domain packages: `data_loader` (data parsing and dividend bootstrapping), `calibration` (Kolmogorov forward PDE solvers and fixed-point algorithms), `pricing` (BS/LV path simulators), `risk_management` (Greeks calculation, weighted L^2 hedging vanilla optimization, P&L attribution), and `reporting` (console and plot formatting).

10.2. Dependency Structure

The system implements a strictly hierarchical, downward dependency flow from orchestrators to leaf functional helpers. A central feature is the shared path simulation Directed Acyclic Graph (DAG) inside `compute_risk.m`. Rather than generating independent paths for each parameter bump and instrument, the orchestrator pre-generates a synchronized Gaussian noise matrix $\mathbf{Z}$ to simulate baseline, spot-bumped, volatility-bumped, and dividend-bumped paths. These are then broadcast to independent contract calculators (`compute_greeks_*.m`), isolating finite-difference sensitivities from Monte Carlo noise through Common Random Numbers (CRN) while avoiding redundant path generation.

10.3. Codebase Statistics

To assess implementation complexity, a `cloc` analysis was performed on the active source tree inside the `src/` directory. Excluding external reports and build directories, the codebase metrics are broken down by component directory below.

Directory / Component	Files	Blank Lines	Comment Lines	Code Lines
<code>src/</code> (Root / Entry points)	2	43	34	152
<code>src/orchestrators/</code> (Execution pipelines)	7	80	113	273
<code>src/helpers/data_loader/</code> (Data loading)	4	11	25	44
<code>src/helpers/calibration/</code> (PDE Calibration)	9	52	108	173
<code>src/helpers/pricing/</code> (MC pricing kernels)	10	41	132	183
<code>src/helpers/risk_management/</code> (Greeks & Hedging)	12	109	204	497
<code>src/helpers/reporting/</code> (Visualization & outputs)	8	95	33	486
Total	52	431	649	1,808

Table 10: Codebase metrics breakdown across the active `src/` directory structure.